

Selective Benchmark — Evaluation Report

Project GitHub Link: <https://github.com/Harshith55072/NeuralDocker-Selective.git>

Data Sheet Used GitHub Link: https://github.com/Harshith55072/DataSheet-s_Of_Projects.git

Project: NeuralDocker-Selective **Evaluated:** 2026-06-11

Judge Model: qwen2.5-1.5b-instruct-q4_k_m **Total Questions:** 300 (100 Factual / 100 Reasoning / 100 Coding)

This test validates the cluster voting (ensemble) system implemented in NeuralDocker-Selective. Four quantised GGUF models were benchmarked individually, then combined via a weighted ensemble. All models ran in a single session with temperature = 0.0, repeat_penalty = 1.1, n_ctx = 2048, max_tokens = 512.

1. Individual Model Results

Model	Quantisation	Overall	Correct / 300	Factual / 100	Reasoning / 100	Coding / 100
gemma-2b-aps-it	Q4_K_M	38.00%	114	54.00%	23.00%	37.00%
Llama-3.2-1B-Instruct	Q4_K_M	83.67%	251	95.00%	69.00%	87.00%
qwen2.5-1.5b-instruct	Q4_K_M	90.00%	270	98.00%	75.00%	97.00%
unsloth (fine-tuned)	Q4_K_M	58.33%	175	86.00%	39.00%	50.00%

Model A — gemma-2b-aps-it-Q4_K_M

Overall: 38.00% (114/300)

Section	Correct	Incorrect	Accuracy
Factual (Q001–Q100)	54	46	54.00%
Reasoning (Q101–Q200)	23	77	23.00%
Coding (Q201–Q300)	37	63	37.00%

Weakest model in the pool. Reasoning accuracy of 23% is notably poor — the model answered fewer than 1 in 4 reasoning questions correctly. Factual performance at 54% is marginally above chance for binary tasks but well below the other models. Coding sits at 37%, roughly in line with its overall average. Total incorrect: 186 / 300.

Model B — Llama-3.2-1B-Instruct-Q4_K_M

Overall: 83.67% (251/300)

Section	Correct	Incorrect	Accuracy
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Factual (Q001–Q100)	95	5	95.00%
Reasoning (Q101–Q200)	69	31	69.00%
Coding (Q201–Q300)	87	13	87.00%

Strong across all three sections. Factual accuracy at 95% is the second-highest recorded. Reasoning (69%) is the model's weak point and represents the largest absolute gap to its best section. Coding at 87% reflects solid performance. Total incorrect: 49 / 300.

Model C — qwen2.5-1.5b-instruct-q4_k_m

Overall: 90.00% (270/300)

Section	Correct	Incorrect	Accuracy
Factual (Q001–Q100)	98	2	98.00%
Reasoning (Q101–Q200)	75	25	75.00%
Coding (Q201–Q300)	97	3	97.00%

Best-performing individual model. Factual (98%) and Coding (97%) are near-ceiling. Reasoning (75%) is its only notable gap. Note: Model C also served as the judge model for all evaluations, introducing potential self-evaluation bias — see limitations. Total incorrect: 30 / 300.

Model D — unsloth.Q4_K_M

Overall: 58.33% (175/300)

Section	Correct	Incorrect	Accuracy
Factual (Q001–Q100)	86	14	86.00%
Reasoning (Q101–Q200)	39	61	39.00%
Coding (Q201–Q300)	50	50	50.00%

Second-weakest. Factual performance (86%) is respectable and the model's clear strength. Reasoning (39%) is extremely poor — on par with near-random selection. Coding at exactly 50% (50/100 correct) is effectively at chance for binary decisions. Total incorrect: 125 / 300.

2. Ensemble Result

Overall: 93.33% (280/300) Evaluated: 2026-06-13

Section	Correct	Incorrect	Accuracy
Factual (Q001–Q100)	98	2	98.00%
Reasoning (Q101–Q200)	84	16	84.00%
Coding (Q201–Q300)	98	2	98.00%

The ensemble (ensemble_ANSWERS) outperforms every individual model. Compared to Model C (the best individual), overall accuracy improved from 90.00% to 93.33% — a gain of +3.33 percentage points. Total incorrect answers dropped from 30 to 20, a reduction of 33.3%.

The largest per-section gain is in Reasoning: from 75.00% (Model C) to 84.00% (ensemble), a +9.00 pp improvement. Factual and Coding were already near-ceiling at the individual level (98% and 97%) and held at 98% in the ensemble.

	Best Individual (Model C)	Ensemble	Delta
Overall	90.00% (270/300)	93.33% (280/300)	+3.33 pp
Factual	98.00% (98/100)	98.00% (98/100)	0.00 pp
Reasoning	75.00% (75/100)	84.00% (84/100)	+9.00 pp
Coding	97.00% (97/100)	98.00% (98/100)	+1.00 pp
Incorrect	30	20	-10 (-33.3%)

3. Weighted Voting Mechanism

The ensemble uses a weighted peer-review voting pipeline (v3). The core idea is that each model votes on the other models' answers, but each vote is scaled by the reviewer's proven accuracy in that question's section. A model with 97% Coding accuracy carries more weight when judging Coding answers than a model with 39% accuracy.

3.1 Section Classification

Questions are bucketed by ID: Q001–Q100 = Factual, Q101–Q200 = Reasoning, Q201–Q300 = Coding.

3.2 Weight Assignment

Weights are loaded automatically by parsing each model's grader report. The regex pattern extracts per-section accuracy from the markdown table row: | Section (Qxxx–Qxxx) | total | correct | incorrect | accuracy% |. The parsed percentage is divided by 100 to produce a weight in [0.0, 1.0]. If parsing fails, weight defaults to 0.5.

Weights used in this run (derived from individual benchmark scores above):

Model	Factual Weight	Reasoning Weight	Coding Weight
gemma-2b-aps-it	0.54	0.23	0.37
Llama-3.2-1B-Instruct	0.95	0.69	0.87
qwen2.5-1.5b-instruct	0.98	0.75	0.97
unsloth.Q4_K_M	0.86	0.39	0.50

3.3 Pipeline Phases

Phase 1 — Answer generation. All 4 models generate an answer independently using temperature = 0.0 (deterministic). Answers are screened for system-prompt leakage via regex; flagged answers are disqualified and receive score 0.

Phase 2 — Peer review. Each model reviews every other model's answer (no self-review). The reviewer assigns a score from 1 to 5 using the scale: 1 = completely wrong, 3 = partially correct, 5 = accurate and complete. The reviewer is constrained to output a single integer via `max_tokens = 8`.

Phase 3 — Weighted tally. The weighted score for each answerer is computed as:

$$\text{weighted_score}(\text{answerer}) = \text{SUM over all reviewers of } [\text{raw_score}(\text{reviewer}) \times \text{weight}(\text{reviewer}, \text{section})]$$

where `weight(reviewer, section)` is the reviewer's proven accuracy fraction for that section. The answerer with the highest `weighted_score` is selected as the winner. Ties are broken by answer length (longer answer wins).

3.4 Disqualification

An answer is disqualified if it matches any leak pattern in its first 200 characters — for example, echoing the system prompt verbatim or containing known prompt-format artifacts. Disqualified answers are scored 0 in the tally. If all answers for a question are disqualified, the winner falls back to the highest-scoring answer regardless of disqualification.

4. Shared Incorrect Questions

Questions that every individual model got wrong are the hardest cases — questions the ensemble must resolve purely through weighting, since no model answered correctly. Conversely, questions that only one model got wrong are the easiest wins for the ensemble.

Common incorrect questions across all 4 models (Q-IDs appearing in all four incorrect lists): Q037, Q103, Q108, Q153, Q182, Q189, Q191. These 7 questions represent the hardest subset in the benchmark. The ensemble resolved some of these through weighted selection but not all — they account for a portion of the ensemble's 20 remaining errors.

Questions that the ensemble answered correctly but the best individual (Model C) did not (i.e., net new correct answers from the ensemble): the ensemble gained 10 answers over Model C. The Reasoning section accounts for 9 of those 10 gains (ensemble Reasoning: 84 vs Model C Reasoning: 75). The ensemble's Reasoning improvement is the primary driver of the overall accuracy gain.

5. Limitations

1. Single-session results. All benchmarks were run in one session. No cross-session variance or confidence intervals are available. Results may differ across runs even at temperature = 0.0 due to batch order effects in llama.cpp.
2. Judge-model bias. The judge model (qwen2.5-1.5b-instruct-q4_k_m) is identical to Model C. This creates a self-evaluation conflict: Model C grades its own answers indirectly through the grading pipeline, and the grader's definition of correctness is aligned with its own output style. Model C's 90% individual score and 0.98 weight in Factual/Coding sections should be interpreted with this in mind.
3. Weighted scoring inherits individual errors. If the judge model mis-scores a question during individual benchmarking, that error propagates into the weight matrix used by the ensemble. Incorrect weights reduce the ensemble's ability to select the best answer.
4. Q4_K_M quantisation applies to all models. Comparison is fair within this experiment but results are not directly comparable to full-precision or differently-quantised versions of the same model families.

6. Summary

Entity	Overall Accuracy	Incorrect / 300	Key Gap
Model A (gemma-2b-aps-it)	38.00%	186	Reasoning: 23%
Model B (Llama-3.2-1B-Instruct)	83.67%	49	Reasoning: 69%
Model C (qwen2.5-1.5b-instruct)	90.00%	30	Reasoning: 75%
Model D (unsloth)	58.33%	125	Reasoning: 39%
Ensemble	93.33%	20	Reasoning: 84%

The weighted ensemble outperforms every individual model. The gain is concentrated in Reasoning (+9.00 pp over the best individual), which is the structurally hardest section and the one where weaker models most reliably contributed noise rather than signal. The judge-model conflict is the primary caveat to these results and should be addressed in future evaluations by using an independent grader.

Incorrect Q-IDs — Ensemble: Q037, Q068, Q108, Q109, Q121, Q127, Q133, Q137, Q141, Q152, Q153, Q176, Q182, Q183, Q189, Q190, Q191, Q194, Q221, Q296